CS/E-111

ARED GROUP INC.

Modular AI System for Real-Time Video Analytics on Edge Devices

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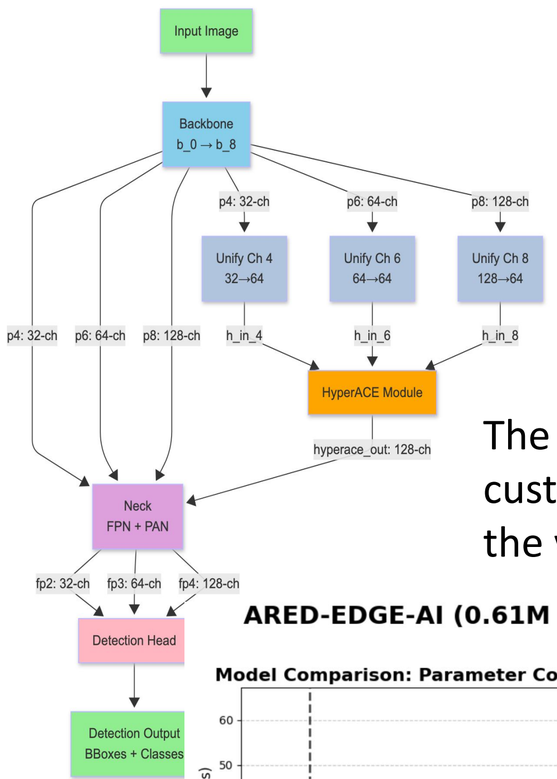
Project Requirements

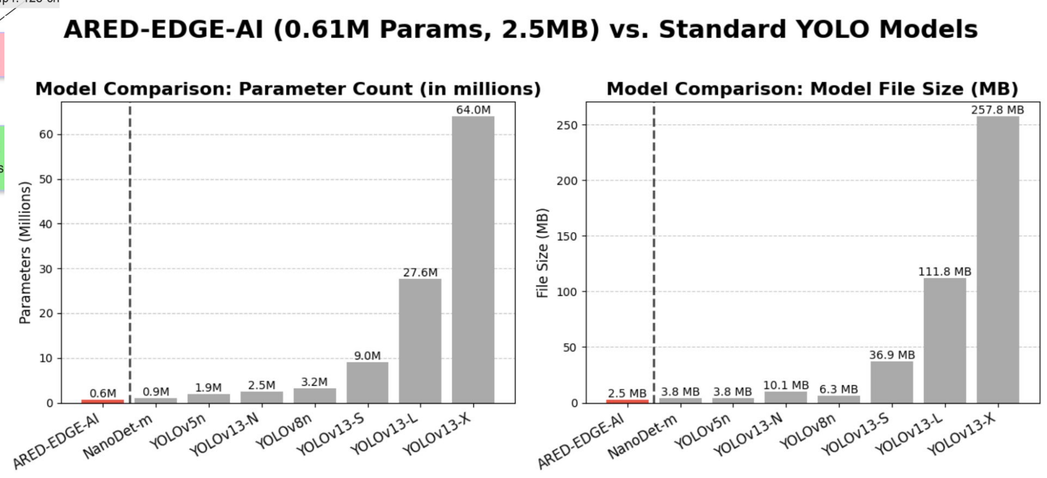
Develop a **modular AI system** for real-time video analytics. With support for multiple **Computer Vision** tasks in a single end-to-end containerized pipeline. Ensure stable performance on edge devices with **limited compute**, and create a connected frontend which allows user control over AI Camera system.

AI Modules

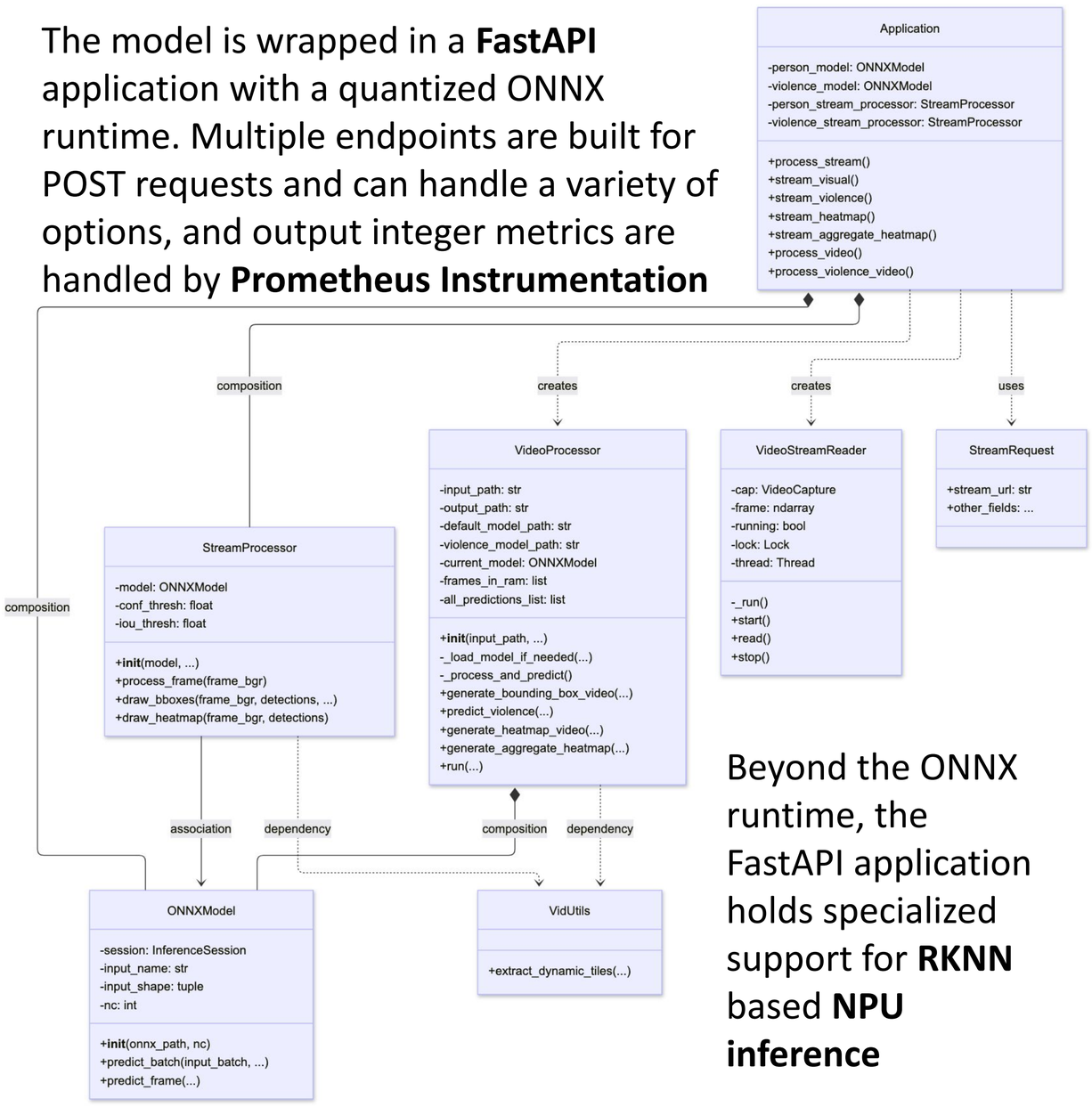


The general-purpose people detection model is trained on COCO 2017 with an architecture based on YOLOv13. The model reached a mAP of **62%**, which is competitive with YOLOv8, but **5.3x** smaller in weights.

The model is then fine-tuned on a custom Roboflow violence dataset for the violence detection module



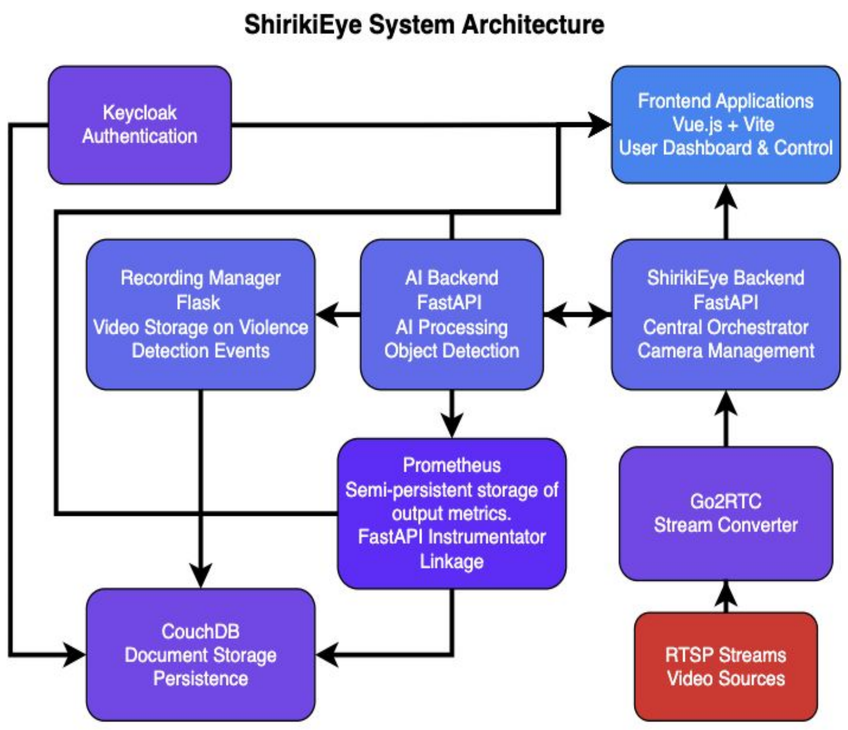
The model is wrapped in a **FastAPI** application with a quantized ONNX runtime. Multiple endpoints are built for POST requests and can handle a variety of options, and output integer metrics are handled by **Prometheus Instrumentation**



Beyond the ONNX runtime, the FastAPI application holds specialized support for **RKNN** based **NPU inference**

Application Description

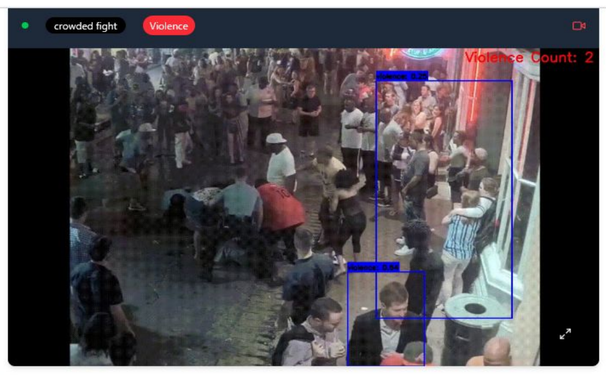
Customers will access the system through a dashboard that displays **live video streams** and **AI outputs** in real-time. Inference types: person count, violence detection, and queue detection, can be toggled directly from the interface. The backend updates the dashboard continuously with annotations and analytics generated by AI modules.

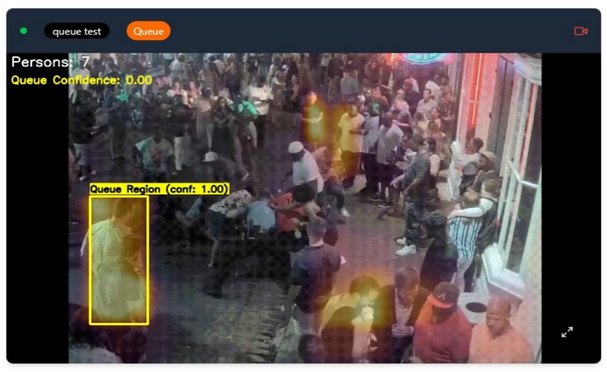


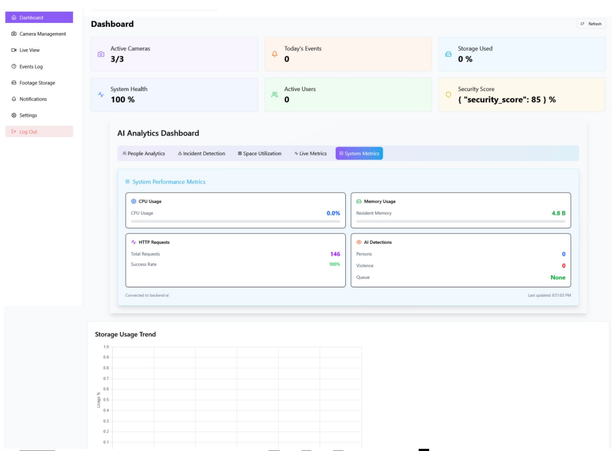
The ShirikiEye System has support systems of databases and authentication systems in order to help backend services run the models, save results, and display metrics on a GUI.

Results

The AI pipeline achieved real-time inference across all tasks, running stably on **CPU/NPU** hardware. Modules produced **low-latency** outputs under containerization, and the dashboard updated detections/analytics reliably in real time. System confirmed consistent resource usage and inference timing.







Future Work

- Add new analytics modules
- Improve model accuracy
- Accelerate Inference
- Expand Dashboard
- Enhance Security Features
- Reduce RAM overhead by replacing Prometheus with a custom instrumentation

RKNN

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